ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION â€" DECEMBER / JUNE 2024

B.TECH. CE ENGINEERING â€" SEMESTER 4

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted. Mobile phones are strictly prohibited.

--- UNIT I ---

Q.1 (a) Differentiate between Static Indeterminacy and Kinematic Indeterminacy. Calculate degree of indeterminacy for given frames and trusses. [7 Marks]
(b) State and prove Castigliano's First and Second Theorems of strain energy. [7 Marks]

[OR]

Q.1 (Alt) Alternative: (a) State Principle of Virtual Work (Unit Load Method) and apply it to calculate deflection at free end of cantilever beam. [7 Marks]
(b) State Betti's Law and Maxwell's Reciprocal Theorem with illustrative proofs. [7 Marks]

--- UNIT II ---

Q.2 (a) Analyze a propped cantilever beam of span L subjected to uniformly distributed load w per unit length using strain energy method. [7 Marks]
(b) State and prove Theorem of Three Moments (Clapeyron's Equation). [7 Marks]

[OR]

Q.2 (Alt) Alternative: Analyze a two-span continuous beam ABC (AB = 6m, BC = 4m) with point load 40 kN at center of AB and UDL 20 kN/m on BC using Clapeyron's Theorem. Draw SFD and BMD. [14 Marks]

--- UNIT III ---

Q.3 (a) Explain Slope Deflection Method. Derive the standard slope deflection equations for members with and without settlement of supports. [7 Marks]
(b) Analyze a continuous beam using Slope Deflection Method and draw Bending Moment Diagram. [7 Marks]

[OR]

Q.3 (Alt) Alternative: Analyze a single-bay, single-story portal frame ABCD with fixed bases subjected to lateral load 30 kN at column joint B using Slope Deflection Method. [14 Marks]

--- UNIT IV ---

Q.4 (a) Explain Moment Distribution Method (Hardy Cross Method). Define Carry-over factor, Stiffness factor, and Distribution factor. [7 Marks]
(b) Analyze continuous beam ABCD with overhang using Moment Distribution Method and locate point of contraflexure. [7 Marks]

[OR]

Q.4 (Alt) Alternative: Analyze symmetrical portal frame with sway using Moment Distribution Method. [14 Marks]

--- UNIT V ---

Q.5 (a) Explain Rolling Loads. Draw influence line diagrams (ILD) for reaction, shear force, and bending moment at a section of simply supported beam. [7 Marks]
(b) State Müller-Breslau Principle and demonstrate how it is used to obtain qualitative ILD for indeterminate structures. [7 Marks]

[OR]

Q.5 (Alt) Alternative: (a) A train of 4 wheel loads 50 kN, 100 kN, 100 kN, and 80 kN spaced 2m apart crosses a girder of span 25m. Calculate absolute maximum bending moment. [7 Marks]
(b) Describe analysis of Three-Hinged Arches and calculate horizontal thrust for parabolic arch carrying UDL over full span. [7 Marks]

**** END OF QUESTION PAPER â€" MITS GWALIOR AUTONOMOUS EXAMINATION SYSTEM ****